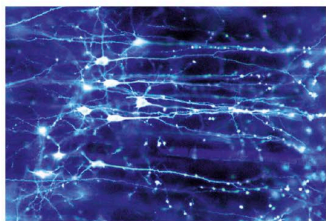


NERVES AND NEURONS

NERVE CELLS, OR NEURONS, ARE HIGHLY SPECIALIZED IN THEIR STRUCTURE, FUNCTION, AND COMMUNICATION LINKS.

NEURON STRUCTURE

Like all other cells, a typical neuron has a main cell body with a nucleus. But a neuron also has long, wire-like processes that connect the neuron to others, allowing messages to be passed at junctions called synapses. These processes are of two main kinds. Dendrites receive messages from other neurons, or from nerve-like cells in sense organs, and conduct them towards the cell body of the neuron. The axon conveys messages away from the cell body, to other neurons or to muscle or gland cells. Dendrites tend to be short and have many branches, while axons are usually longer and branch less along their length. Neurons in the brain and spinal cord are protected and nurtured by supporting nerve cells known as glial cells.



MICROSCOPE VIEW

Nerve cells under the microscope display their cell bodies with nuclei (left) and processes (right).

Mitochondrion
Involved in cell respiration and production of energy

Nucleus
Located towards the middle of the cell body

Cell body

Axon terminal fibre

Schwann cell
Produces myelin

Schwann cell nucleus

Dendrite process
Receives messages from other neurons

Axon process
Transmits messages from the nerve cell body to other tissues

